

Barkot Tamrat Simegn - Click to Download

barkotsimegn@gmail.com • 720-903-6855 • <https://barkotsimegn.github.io/barkot-simegn>

EDUCATION

Northwestern University | Evanston, IL

June 2027

Bachelor of Science in Mechanical Engineering GPA 3.3/4.0

Community College of Aurora | Aurora, CO

May 2023

A.S. in Physics GPA: 3.8/4.0

Overland High School | Aurora, CO GPA: 4.0/4.0

May 2023

Relevant Courses: Material Science, Mechanics of Materials, Thermodynamics, Differential Equations, Fluid Dynamics, Statics, Dynamics, Multivariable Calculus, Scientific and Embedded Programming in Python, Electronics Design, Linear Algebra

SKILLS

Technical Skills: Prototyping, Python, Circuit Python, Matlab, Breadboarding, Microcontrollers, CAD, CAM, Certified Solidworks Professional, Certified Solidworks Associate, Microsoft Suite, Soldering, Excel, HTML/CSS, Spanish

PROJECTS

Northwestern Underwater Robotics Frame Team

Sep 2024 - Present

- Design and build a brand new frame design for an underwater robot using Solidworks and Onshape.
- Manufactured using a variety of methods including water jetting HPV, milling brackets, and 3d printing fan covers.
- Assembled and tested an underwater robot frame and iterated upon the design for a competition.

Modified Bike Stand|Shirley Ryan Ability Lab

March - June 2024

- Collaborated with a multidisciplinary team to create a bike stand for a student with cerebral palsy, enabling independent use and increased accessibility.
- Conducted research, user interviews, and data collection to inform design decisions to meet specific accessibility needs.
- Utilized Solidworks for computer-aided design and 3D printing technology to create and test functional prototypes.

Northwestern NSBE Finance Chair|McCormick School of Engineering

Jan-April 2024

- Managed finances for the Northwestern University chapter of the National Society of Black Engineers (NSBE), overseeing fundraising initiatives and chapter events.
- Coordinated chapter meetings and events, ensuring effective communication and collaboration among team members.
- Gained leadership experience through event management, financial oversight, and community engagement.

Ostomy Bag Cleaner|Shirley Ryan Ability Lab

Sep - Dec 2023

- Designed a compassionate and functional solution for cleaning ostomy bags, improving user independence and quality of life for patients.
- Conducted primary research, including interviews with patients and medical professionals, to understand user needs and design requirements.
- Utilized Solidworks for prototyping and 3D printing to develop and test a working model, culminating in a viable product ready for implementation.

PROFESSIONAL EXPERIENCE

Manager|City of Aurora|Aurora CO

May 2022 - Present

- Supervised daily operations at various city facilities, ensuring patron safety, facility maintenance, and operational efficiency.
- Directed many different teams of staff responsible for maintaining facility cleanliness, executing routine safety inspections, and enforcing safety protocols.
- Oversaw dozens of training sessions to ensure team members were up-to-date on safety regulations and operational procedures.
- Promoted from a temporary position to Manager on Duty, demonstrating leadership, conflict resolution, and team management skills through consistent performance.

Cardboard Carnival|Y.O.U. Maker Space|Evanston IL

Jan-April 2024

- Orchestrated educational workshops for underrepresented youth, introducing foundational concepts in engineering, design, and coding.
- Taught a group of 20 students how to write basic code to control servo motors, fostering problem-solving skills and creativity through hands-on projects.
- Mentored students in collaborative design thinking, encouraging teamwork and iterative development processes.
- Gave back to the community by inspiring and empowering youth to explore STEM.